

MCQS

CS614 midterm preparation file

ZB

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

In the name of God , Most
Gracious, Most merciful



CS614

MCQS

1. What is the primary goal of a data warehouse?

- a) Store raw data
- b) Perform transactions
- c) Enable decision-making
- d) Maintain system logs

Answer: c) Enable decision-making

2. In data warehousing, the term "ETL" stands for:

- a) Extract, Transform, Load
- b) Export, Transfer, Load
- c) Extract, Transfer, List
- d) Extract, Track, Link

Answer: a) Extract, Transform, Load

3. A key characteristic of a data warehouse is:

- a) Volatile
- b) Integrated
- c) Decentralized
- d) Distributed

Answer: b) Integrated

4. What does OLAP stand for?

- a) Online Analytical Processing
- b) Online Application Processing
- c) Online Access Processing
- d) Online Array Processing

Answer: a) Online Analytical Processing

5. Which schema is most commonly used in data warehousing?

- a) Star schema
- b) Network schema
- c) Hierarchical schema
- d) Flat schema

Answer: a) Star schema

6. The fact table in a star schema contains:

- a) Only dimensions
- b) Measures and keys
- c) Metadata
- d) Only primary keys

Answer: b) Measures and keys

7. A data mart is typically:

- a) A large-scale warehouse
- b) Department-specific
- c) For transactional systems
- d) Highly normalized

Answer: b) Department-specific

8. Which of the following is a benefit of data warehousing?

- a) Slow query processing
- b) Improved decision-making
- c) Reduced data quality
- d) Complex maintenance

Answer: b) Improved decision-making

9. Which type of OLAP stores data in relational databases?

- a) MOLAP
- b) HOLAP
- c) ROLAP
- d) DOLAP

Answer: c) ROLAP

10. In a star schema, dimension tables are:

- a) Highly normalized
- b) Highly denormalized
- c) Unrelated to the fact table
- d) Stored in flat files

Answer: b) Highly denormalized

11. The term "granularity" in a data warehouse refers to:

- a) The size of data blocks
- b) The level of detail in the data
- c) The number of dimensions
- d) The speed of query execution

Answer: b) The level of detail in the data

12. Which data model allows multiple fact tables?

- a) Star schema
- b) Snowflake schema
- c) Fact constellation schema
- d) None of the above

Answer: c) Fact constellation schema

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13. Metadata in a data warehouse is used for:

- a) Storing raw data
- b) Data extraction
- c) Describing data
- d) Query execution

Answer: c) Describing data

14. Data cleansing involves:

- a) Deleting old data
- b) Correcting errors in data
- c) Duplicating data
- d) Storing data in the cloud

Answer: b) Correcting errors in data

15. Which phase of ETL involves removing duplicates?

- a) Extraction
- b) Transformation
- c) Loading
- d) Indexing

Answer: b) Transformation

16. Which is NOT a type of OLAP operation?

- a) Roll-up
- b) Drill-down
- c) Slice
- d) Sorting

Answer: d) Sorting

17. MOLAP typically stores data in:

- a) Relational tables
- b) Multidimensional arrays
- c) CSV files
- d) Text files

Answer: b) Multidimensional arrays

18. Data warehouses are optimized for:

- a) Transactional processing
- b) Analytical queries
- c) Real-time updates
- d) Operational efficiency

Answer: b) Analytical queries

19. A snowflake schema differs from a star schema in:

- a) Centralized fact tables
- b) Denormalized dimensions
- c) Normalized dimensions
- d) Fewer relationships

Answer: c) Normalized dimensions

20. Which of the following is NOT a characteristic of a data warehouse?

- a) Time-variant
- b) Non-volatile
- c) Subject-oriented
- d) Highly volatile

Answer: d) Highly volatile

21. What is a "drill-down" operation in OLAP?

- a) Aggregating data
- b) Moving to a higher level of detail
- c) Removing unnecessary data
- d) Merging data sets

Answer: b) Moving to a higher level of detail

22. Which component of a data warehouse ensures data consistency?

- a) Data staging
- b) Metadata repository
- c) ETL tools
- d) OLAP tools

Answer: c) ETL tools

23. What is the purpose of data transformation in ETL?

- a) Cleaning and formatting data
- b) Moving data to a warehouse
- c) Generating reports
- d) Archiving old data

Answer: a) Cleaning and formatting data

24. A data warehouse is updated:

- a) Continuously
- b) In real-time
- c) Periodically
- d) Upon user request

Answer: c) Periodically

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25. Which of these is NOT a layer in a data warehouse architecture?

- a) Data source layer
- b) Data staging layer
- c) User interface layer
- d) Buffer layer

Answer: d) Buffer layer

26. Star schema improves:

- a) Query complexity
- b) Query performance
- c) Data redundancy
- d) Data normalization

Answer: b) Query performance

27. Which type of OLAP combines ROLAP and MOLAP?

- a) DOLAP
- b) HOLAP
- c) COLAP
- d) VOLAP

Answer: b) HOLAP

28. The "slice" operation in OLAP is used for:

- a) Filtering data on one dimension
- b) Aggregating data across all dimensions
- c) Comparing two datasets
- d) Normalizing data

Answer: a) Filtering data on one dimension

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29. Which of the following stores historical data for analysis?

- a) OLAP systems
- b) Transactional databases
- c) Data warehouses
- d) Operational systems

Answer: c) Data warehouses

30. What is a dimension in data warehousing?

- a) A measure of performance
- b) A perspective for analysis
- c) A transactional log
- d) A data cleansing tool

Answer: b) A perspective for analysis

31. Surrogate keys in a data warehouse are:

- a) Natural keys
- b) System-generated keys
- c) Foreign keys
- d) Composite keys

Answer: b) System-generated keys

32. A data cube in OLAP is:

- a) A graphical tool
- b) A multi-dimensional data structure
- c) A collection of metadata
- d) A normalized table

Answer: b) A multi-dimensional data structure

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33. Which type of data warehouse architecture uses middleware to integrate multiple sources?

- a) Virtual**
- b) Federated**
- c) Centralized**
- d) Distributed**

Answer: b) Federated

34. What is the main advantage of a snowflake schema over a star schema?

- a) Faster queries**
- b) Reduced redundancy**
- c) Easier to implement**
- d) Denormalized structure**

Answer: b) Reduced redundancy

35. What does the "Load" phase in ETL involve?

- a) Transforming data**
- b) Storing data in the target system**
- c) Extracting data from sources**
- d) Cleaning and formatting data**

Answer: b) Storing data in the target system

36. Which type of OLAP tool is most suitable for low storage requirements?

- a) ROLAP**
- b) MOLAP**
- c) HOLAP**
- d) DOLAP**

Answer: b) MOLAP

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37. Which of the following is an example of a derived attribute in a data warehouse?

- a) Customer name
- b) Order ID
- c) Total sales
- d) Product code

Answer: c) Total sales

38. The main advantage of using surrogate keys in a data warehouse is:

- a) Reduced query time
- b) Improved data consistency
- c) Compatibility with external systems
- d) Reduced storage requirements

Answer: b) Improved data consistency

39. Which type of data is commonly used in data warehouses?

- a) Real-time data
- b) Historical data
- c) Sensor data
- d) Streaming data

Answer: b) Historical data

40. A "roll-up" operation in OLAP results in:

- a) Aggregated data
- b) Detailed data
- c) Filtered data
- d) Normalized data

Answer: a) Aggregated data

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41. What is a key challenge in data integration during ETL?

- a) High-speed querying
- b) Handling data heterogeneity
- c) Performing drill-down operations
- d) Designing user interfaces

Answer: b) Handling data heterogeneity

42. Which data warehouse schema uses hierarchies within dimensions?

- a) Star schema
- b) Snowflake schema
- c) Flat schema
- d) Data vault

Answer: b) Snowflake schema

43. Which of the following is NOT an OLAP operation?

- a) Pivot
- b) Roll-up
- c) Drill-through
- d) Replication

Answer: d) Replication

44. Which term refers to the process of storing detailed data in a data warehouse?

- a) Data summarization
- b) Data granularity
- c) Data partitioning
- d) Data drilling

Answer: b) Data granularity

45. Data latency in a data warehouse refers to:

- a) The time required for query execution
- b) The delay in loading data
- c) The speed of data extraction
- d) The frequency of system downtime

Answer: b) The delay in loading data

46. A cube in OLAP is defined by its:

- a) Dimensions and measures
- b) Rows and columns
- c) Queries and indices
- d) Keys and attributes

Answer: a) Dimensions and measures

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WhatsApp:

<https://whatsapp.com/channel/0029VaODheTDp2Q8YVco0N2p>

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channel on WhatsApp:**

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.

Therefore, for passing a course/subject, student shall fulfill the following:

- Secure minimum **20%** score in Formative Assessments/Mid terms
- Secure minimum **20%** score in Final Term Examinations.
- Secure at least **40%** marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?

A: Follow these steps:

- 1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).**
- 2. Each subject typically has 3 credit hours.**
- 3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.**

Examples:

21 credit hours = 7 subjects

15 credit hours = 5 subjects

Remember, the university may allow different credit hours for each student, so check your specific allowance.

By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed

Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish
krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat
Fame

Moat

**Humy pta hona chaheay Moat tab ani ha jab Allah
chahy us se pehly puri dunya bi ak taraf ho ke apko
marna chahy to apka kuch nahe beggar saqti**

Risk/Dolat

**Risk Dolat sb Allah pak ke hath m ha puri dunya ak
taraf b ho jay na apsy apka risk cheen saqti ha na de
saqti ha Ye sirf ALLAH pak ky hath m ha**

Izat Zilat shohrat Fame

**Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b
apko zra brabr b damage nahe kr saqti Agr ALLAH**

**Apko izat dena chahay to puri dunya ak taraf ho ke
b apko 1% b nuksan nahe pohancha saqti**

Phr Dar ks bat ka ????????????????

**Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne
mat jhuko puri dunya mel k b apka kuch b nahe
beggar saqti**

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family

Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my campaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha

Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us

m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us

Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

..... **MY Family**

**ust Relax and focus! Exams are not difficult. Put in your
100% effort and trust in Allah.**

..... **AL-Quran**

**"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"**

May Allah bless you